

**a) Problem: Find out the n'th fibonacci number.**

**Logical Solution:**

- Fibonacci pattern can be generated by the summation of previous 2 numbers where first 2 numbers are fixed which are 0,1.
- $A_n = A_{n-1} + A_{n-2}$

**Source Code:**

```
def fibonacci(n):
    if n==0:
        return 0
    if n<=2:
        return 1
    if value[n]!=0:
        return value[n]
    value[n]=fibonacci(n-1)+fibonacci(n-2)
    return value[n]

n=int(input("Enter a value: "))
value=[0]*(n+1)
print(f'{n} no. fibonacci number is {fibonacci(n)}')
```

**Output:**

```
Enter a value: 9
9 no. fibonacci number is 34
PS F:\RUET CSE LAB\3210-Digital Signal Processing\1-17.09.2025>
```

**Discussion:**

- Here I have used recursive and Dynamic approach to solve the problem.

**b) Problem: Take Student ID as Data and do some statistical operation on it.**

**Logical Solution:**

- For sum = summation of every digit of the ID.

- For mean = sum / (Total numbers of digits)
- For variance =  $\sum \frac{(digit - mean)^2}{total\ digits}$

### Source Code:

```
import numpy as np
import matplotlib.pyplot as plt

roll="2103064"
data=np.array([int(x) for x in roll])
print(f'Elements of data: {data}')
print(f'Summation of the data: {np.sum(data)}')
print(f'Mean of the data: {(np.mean(data)).4}')
print(f'Variance of the data: {np.var(data):.4}')

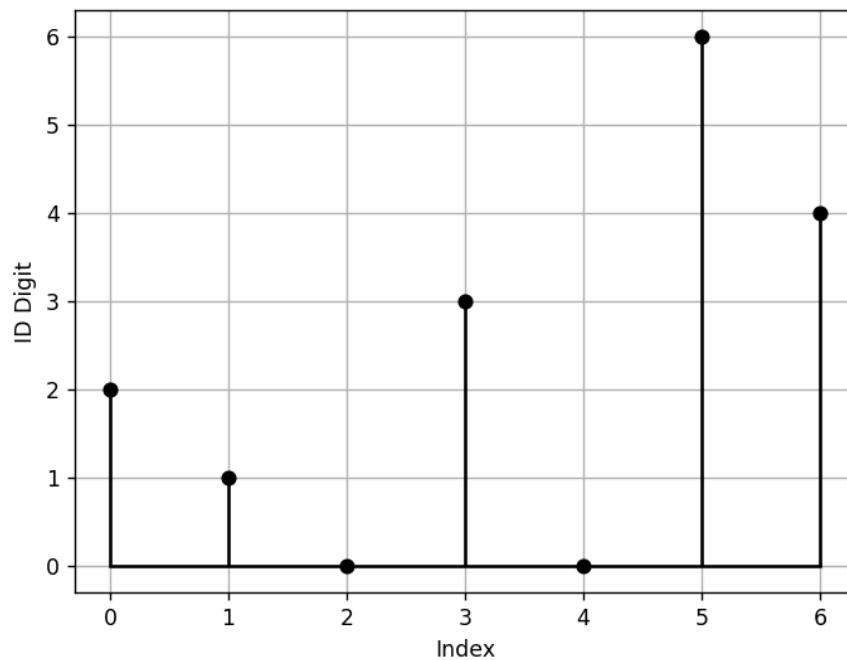
sq_data=data**2
print(sq_data)

markerline,stemlines,baseline=plt.stem(data)

plt.setp(markerline, color='black')
plt.setp(baseline, color='black')
plt.setp(stemlines, color='black')
plt.xlabel("Index")
plt.ylabel("ID Digit")
plt.grid(True)
plt.show()
```

### Output:

```
Elements of data: [2 1 0 3 0 6 4]
Summation of the data: 16
Mean of the data: 2.286
Variance of the data: 4.204
[ 4  1  0  9  0 36 16]
```



**Discussion:**

- As I have used numpy module, I have done the calculation of mean, sum, variance using numpy built-in functions.
- List comprehension is used to store the digits.
- Use matplotlib package for displaying the graph.
- For getting the discrete value, stem() function is used.

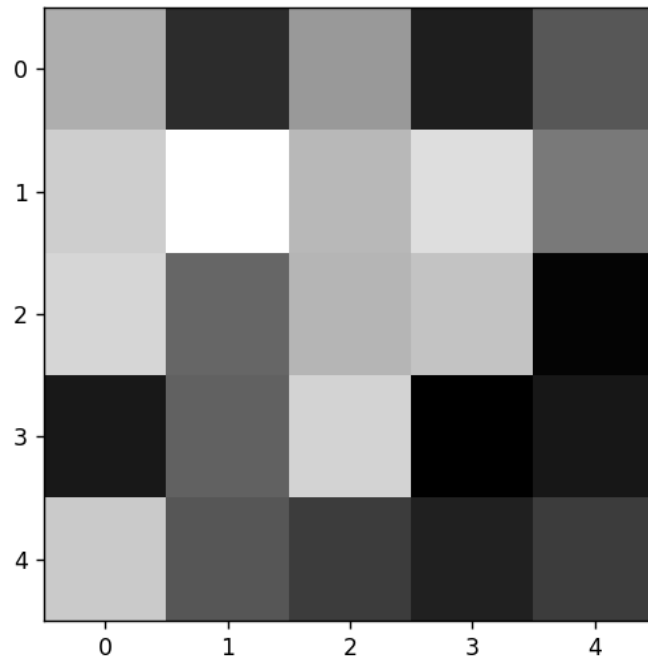
**c) Problem: create Image Pixel Array of 5/5 with random number between 0 to 255.**

**Source Code:**

```
import numpy as np
import matplotlib.pyplot as plt

image=np.random.randint(0,255,size=(5,5))
plt.imshow(image,cmap='gray')
plt.show()
```

### Output:



### Discussion:

- Matplotlib package is used to draw the image pixel
- `cmap='gray'` is used to get the black and white image.
- Random function helps to create 0 to 255 integers randomly.

### d) Problem: Compare the growth rates of the equations- linear, quadratic and exponential.

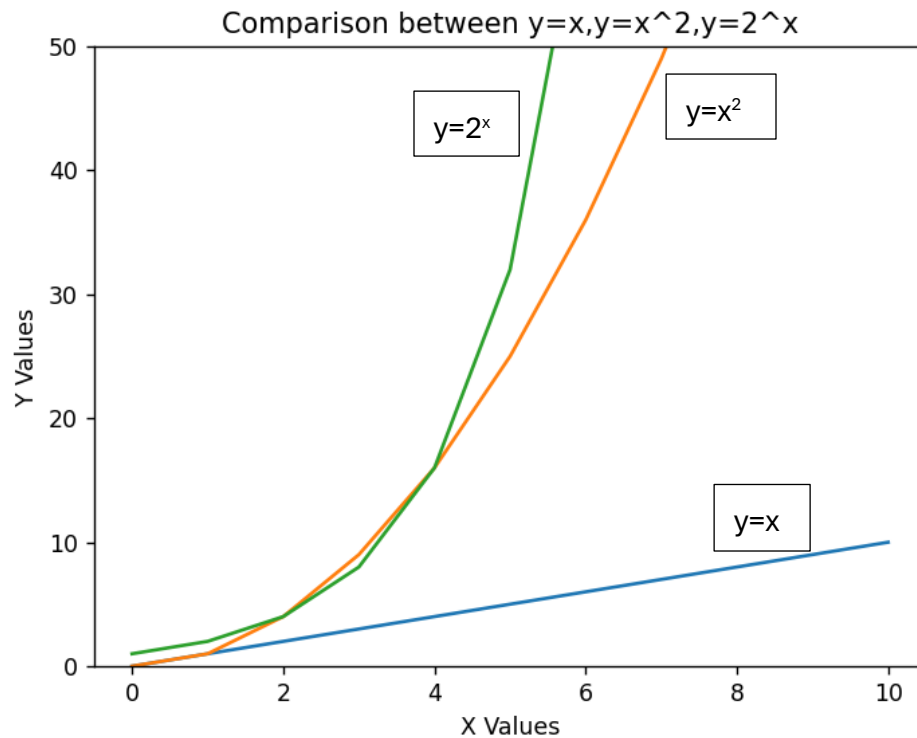
#### Source Code:

```
import numpy as np
import matplotlib.pyplot as plt

x=np.array([x for x in range(11)])
y1=x
y2=x**2
y3=2**x
plt.title("Comparison between y=x,y=x^2,y=2^x")
plt.xlabel("X Values")
plt.ylabel("Y Values")
```

```
plt.plot(x,y1)
plt.plot(x,y2)
plt.plot(x,y3)
plt.ylim(0,50)
plt.show()
```

**Output:**



**Discussion:**

- List comprehension is used to store the x-values.
- Plot the 3 equations using plot() function 3 times.
- From the graph it is clear that  $y=2^x$  dominates the remaining 2 equations.
- To get all the graphs y-value is limited by the margin of 50.

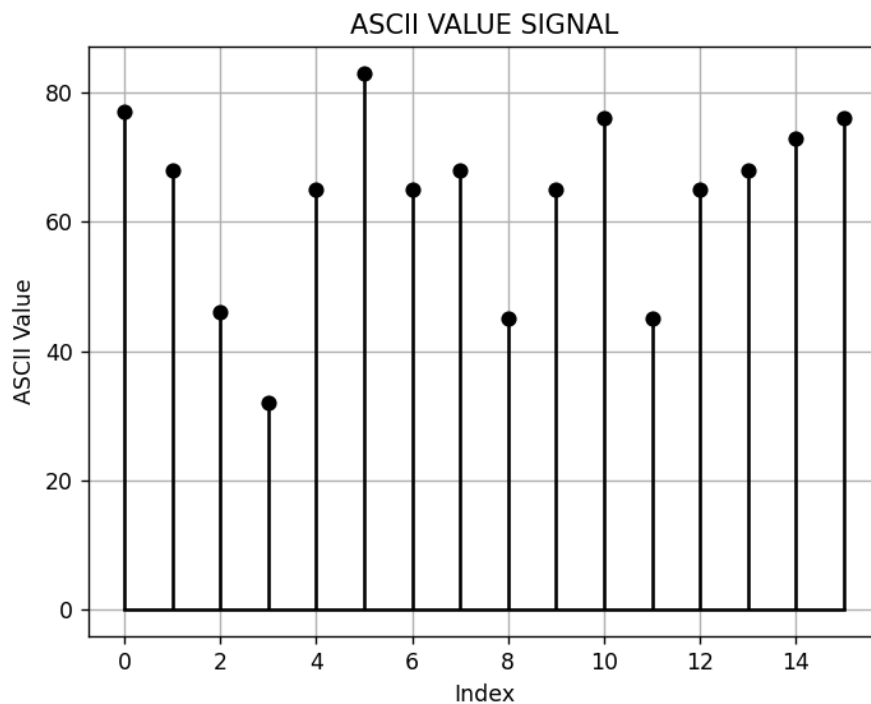
**e) Problem: Encode Your Name as Signal and plot it into the graph.**

**Source Code:**

```
import numpy as np
import matplotlib.pyplot as plt
```

```
name="MD. ASAD-AL-ADIL"  
name=np.array([ord(x) for x in name])  
  
markerline,stemlines,baseline=plt.stem(name)  
plt.setp(markerline, color='black')  
plt.setp(baseline, color='black')  
plt.setp(stemlines, color='black')  
  
plt.title("ASCII VALUE SIGNAL")  
plt.xlabel("Index")  
plt.ylabel("ASCII Value")  
plt.grid(True)  
plt.show()
```

### Output:



### Discussion:

- To get the ASCII value from the name of each character, I have used `ord()` function and list comprehension to store the value.
- To get the discrete graph, `stem()` function is used.